Statistics

Chapter 13 - Hinglish Notes, Mean/Median/Mode + Solutions

Ye Chapter 13 (Statistics) ka complete guide hai. Pehle red-page ke core basics (data, mean/median/mode, class interval, class mark, cf), fir chapter ka main content - mean ke 3 methods, mode aur median ke formulas, aur ogive. End me solved examples aur 20 practice questions ke step-by-step solutions hain.

EXAM * aise green tag wale topics exams me baar-baar aate hain

PART A - Core Basics (Miss Mat Karna)

1. Data

EXAM * Exam me CONSTANTLY aata hai

- Data = collect kiye gaye numbers / observations.
- Grouped data = intervals me bata hua (0-10, 10-20).

2. Mean, Median, Mode (Basic Meaning)

EXAM * Exam me CONSTANTLY aata hai

- Mean = average (saara total / count).
- Median = beech wali value (data sort karke).
- Mode = sabse zyada baar aane wali value.

3. Class Interval aur Frequency

EXAM * Exam me CONSTANTLY aata hai

- Class interval = range (jaise 10-20).
- Frequency (fi) = us interval me kitne items hain.

4. Class Mark (xi)

EXAM * Exam me CONSTANTLY aata hai

- Class mark = interval ka midpoint.
- xi = (upper + lower) / 2.   (0-10 -> 5)

5. Summation (Sigma)

- Sum(fi) = saari frequencies ka total (= n).
- Sum(fi xi) = har (fi x xi) ka total.

6. Average Nikaalna

- Average = total / count. Mean isi par based hai.

7. Cumulative Frequency (cf)

EXAM * Exam me CONSTANTLY aata hai

- cf = frequencies ko add karte jao (running total).
- Median nikaalne ke liye zaroori hai.

PART B - Chapter 13 ka Main Content

8. Mean - Method 1: Direct Method

EXAM *** Exam me CONSTANTLY aata hai**

Mean = Sum(fi xi) / Sum(fi)

xi = class mark, fi = frequency.

9. Mean - Method 2: Assumed Mean Method

EXAM *** Exam me CONSTANTLY aata hai**

Mean = a + [Sum(fi di) / Sum(fi)]

a = assumed mean, di = xi - a.

10. Mean - Method 3: Step Deviation Method

EXAM *** Exam me CONSTANTLY aata hai**

Mean = a + h x [Sum(fi ui) / Sum(fi)]

ui = (xi - a) / h, h = class size.

11. Mode (Grouped Data)

EXAM *** Exam me CONSTANTLY aata hai**

Mode = l + [(f1 - f0) / (2f1 - f0 - f2)] x h

l = modal class ki lower limit, f1 = modal class freq,
f0 = pichli class freq, f2 = agli class freq, h = class size.

12. Median (Grouped Data)

EXAM *** Exam me CONSTANTLY aata hai**

Median = l + [(n/2 - cf) / f] x h

l = median class lower limit, n = Sum(fi),
cf = pichli class tak cumulative freq, f = median class freq, h = class size.

13. Empirical Relation aur Ogive

EXAM *** Exam me CONSTANTLY aata hai**

- Empirical: 3 Median = Mode + 2 Mean.

- Ogive = cumulative frequency curve.
- Less-than aur more-than ogive ka intersection point -> Median.

Quick Revision

> Mean (direct) = Sum(fi xi)/Sum(fi).
> Step deviation: a + h x Sum(fi ui)/Sum(fi).
> Mode = l + [(f1-f0)/(2f1-f0-f2)] x h.
> Median = l + [(n/2 - cf)/f] x h.
> 3 Median = Mode + 2 Mean.

Solved Examples - Samjho Kaise Solve Karte Hain

Neeche 10 examples step-by-step solve karke dikhaye gaye hain.

Example 1: Numbers 4, 6, 8, 10, 12 ka mean?

- Step: Mean = total / count.
- Step: = (4+6+8+10+12)/5 = 40/5.

Answer: Mean = 8

Example 2: 5, 8, 9, 11, 14 ka median?

- Step: Data sorted hai, count = 5 (odd).
- Step: Beech wali (3rd) value lo.

Answer: Median = 9

Example 3: 2, 3, 3, 5, 3, 7 ka mode?

- Step: Sabse zyada baar konsi value? 3 teen baar.

Answer: Mode = 3

Example 4: Class mark of 20-30?

- Step: xi = (upper + lower)/2 = (20+30)/2.

Answer:25

Example 5: Mean=10, Mode=12. Empirical se Median?

- Step: 3 Median = Mode + 2 Mean = 12 + 20 = 32.
- Step: Median = 32/3.

Answer: Median = 10.67

Example 6: 6 numbers, Sum(fi xi)=300, n=6. Mean?

- Step: Mean = Sum(fi xi)/n = 300/6.

Answer: Mean = 50

Example 7: Even numbers 2,4,6,8 ka mean?

- Step: = (2+4+6+8)/4 = 20/4.

Answer:5

Example 8: Data 7,7,8,9,7,10 ka mode?

- Step: 7 sabse zyada (3 baar).

Answer: Mode = 7

Example 9: Median class l=10, n=20, cf=8, f=6, h=10. Median?

- Step: Median = l + [(n/2 - cf)/f] x h.
- Step: = 10 + [(10 - 8)/6] x 10 = 10 + (2/6)x10.

- Step: = 10 + 3.33.

Answer: Median = 13.33

Example 10: Modal class l=20, f1=15, f0=10, f2=12, h=10. Mode?

- Step: Mode = l + [(f1-f0)/(2f1-f0-f2)] x h.
- Step: = 20 + [(15-10)/(30-10-12)] x 10.
- Step: = 20 + (5/8)x10 = 20 + 6.25.

Answer: Mode = 26.25

Practice Questions

Pehle khud solve karne ki koshish karo. Neeche har question ka step-by-step solution diya gaya hai.

Q1. Mean ka matlab kya hota hai?

Q2. Median ka matlab kya hota hai?

Q3. Mode ka matlab kya hota hai?

Q4. Class mark (xi) ka formula kya hai?

Q5. Direct method se mean ka formula kya hai?

Q6. Step deviation method ka formula kya hai?

Q7. Mode (grouped) ka formula kya hai?

Q8. Median (grouped) ka formula kya hai?

Q9. Empirical relation kya hai (mean/median/mode)?

Q10. Cumulative frequency (cf) kya hoti hai?

Q11. 3, 5, 7, 9, 11 ka mean?

Q12. 10, 20, 30, 40, 50 ka median?

Q13. 4, 4, 5, 6, 4, 7 ka mode?

Q14. Class mark of 30-40?

Q15. Mean=15, Mode=18 -> median (empirical)?

Q16. Ogive kya hota hai?

Q17. Do ogive ka intersection point kya deta hai?

Q18. 2, 4, 6, 8, 10 ka mean?

Q19. n=30, n/2 kya hota hai?

Q20. Modal class kise kehte hain?

Step-by-Step Solutions

Q1. Mean ka matlab?

- Step: Total / count.

Answer: Average

Q2. Median ka matlab?

- Step: Sort karke beech wali.

Answer: Beech wali value

Q3. Mode ka matlab?

- Step: Sabse zyada baar.

Answer: Sabse zyada aane wali value

Q4. Class mark formula?

- Step: Midpoint.

Answer:(upper + lower)/2

Q5. Direct mean formula?

- Step: Standard.

Answer: Sum(fi xi)/Sum(fi)

Q6. Step deviation formula?

- Step: Standard.

Answer:a + h x Sum(fi ui)/Sum(fi)

Q7. Mode (grouped) formula?

- Step: Standard.

Answer:l + [(f1-f0)/(2f1-f0-f2)] x h

Q8. Median (grouped) formula?

- Step: Standard.

Answer:l + [(n/2 - cf)/f] x h

Q9. Empirical relation?

- Step: Standard.

Answer:3 Median = Mode + 2 Mean

Q10. cf kya hoti hai?

- Step: Running total.

Answer: Cumulative (running) frequency

Q11. 3,5,7,9,11 mean?

- Step: 35/5.

Answer:7

Q12. 10..50 median?

- Step: Beech (3rd) value.

Answer:30

Q13. 4,4,5,6,4,7 mode?

- Step: 4 teen baar.

Answer:4

Q14. Class mark 30-40?

- Step: (30+40)/2.

Answer:35

Q15. Mean=15, Mode=18 median?

- Step: 3M=18+30=48 -> M=16.

Answer:16

Q16. Ogive kya hota?

- Step: cf curve.

Answer: Cumulative frequency curve

Q17. Do ogive intersection?

- Step: x-coordinate.

Answer: Median

Q18. 2,4,6,8,10 mean?

- Step: 30/5.

Answer:6

Q19. n=30, n/2?

- Step: 30/2.

Answer:15

Q20. Modal class?

- Step: Sabse zyada frequency wali class.

Answer: Highest frequency class